

Low-Level Software Design Document (LLD)

Module Interfaces, Component Hierarchy, State Machines & Pipeline Implementation

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Backend Module Architecture

The Django project core is structured into isolated, cohesive domain packages:

- `core/views_chat.py` : Manages the Ollama IPC bridge, RAG keyword matching, and streaming Server-Sent Events (SSE).
- `ml_churn/` : Encapsulates `train_churn.py` , feature transformers, and serialized `model_lightgbm.pkl` .
- `ml_recommend/` : Implements FunkSVD SGD matrix factorization and cosine collaborative filtering.
- `appliances/` : Handles CRUD and multi-attribute geospatial search across appliances.
- `bookings/` : Enforces the booking state machine transitions.

2. Frontend Component Hierarchy (React 19)

```
[App.jsx (Router & Global Providers)]
|-- [Navbar.jsx (Auth state, Cart badge, Model indicator)]
|-- [Catalog.jsx]
|   |-- [FilterSidebar.jsx (Category, Price slider, Availability)]
|   |-- [ApplianceGrid.jsx]
|       |-- [ApplianceCard.jsx (Image, specs, Rent CTA)]
|-- [AIChatbot.jsx (Floating AI Assistant)]
|   |-- [Header & Status Bar (Model switcher, auto-connect)]
|   |-- [MessageList.jsx (User/AI bubbles, diagnostic cards)]
|   |-- [InputBar.jsx (Voice/text send)]
|-- [AdminDashboard.jsx (BI Churn Monitor, Fleet Logistics)]
```

3. Error Handling & Resilient Degradation

When the Ollama local inference engine is initializing or temporarily busy, `AIChatbot.jsx` renders an interactive **Diagnostic Error Card**. It presents non-technical explanations alongside two one-click recovery triggers: `[ Auto-Start & Retry ]` and `[ Switch to Ultra-Fast Model (qwen2.5:0.5b) ]` , ensuring zero user dead-ends.